Questions by topic: Global climate change - Set 1 (85 marks)

Answer all questions in the spaces provided.

0 1 Table 1 shows information about some greenhouse gases.

Table 1

Gas	Main anthropogenic sources	Approximate residence time	Global warming potential (per molecule, relative to CO_2 over 100 years)
Carbon dioxide	Fossil fuel combustion, deforestation, cement	100+ years (variable)	1
Methane	Rice paddies, cattle, landfill, gas leaks, coal mines	About 12 years	28
Nitrous oxide (NOx)	Fertilisers, combustion	About 120 years	265
CFCs	Refrigerants, aerosols (now banned)	50-100+ years	5 000-10 000
Tropospheric ozone	Secondary pollutant from NOx and hydrocarbons	Days-weeks	-

0 1 . 1 Explain the natural greenhouse effect.

[3 marks]

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0 1 . 2 Use Table 1 to explain why carbon dioxide is considered the most important anthropogenic greenhouse gas even though methane and CFCs absorb far more infrared per molecule.

[3 marks]

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0 1 . 3 Explain why reducing methane emissions would produce a faster reduction in warming than reducing CO_2 emissions by the same mass.

[2 marks]

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0 1 . 4 Give **two** methods of reducing methane emissions from agriculture or waste.

[2 marks]

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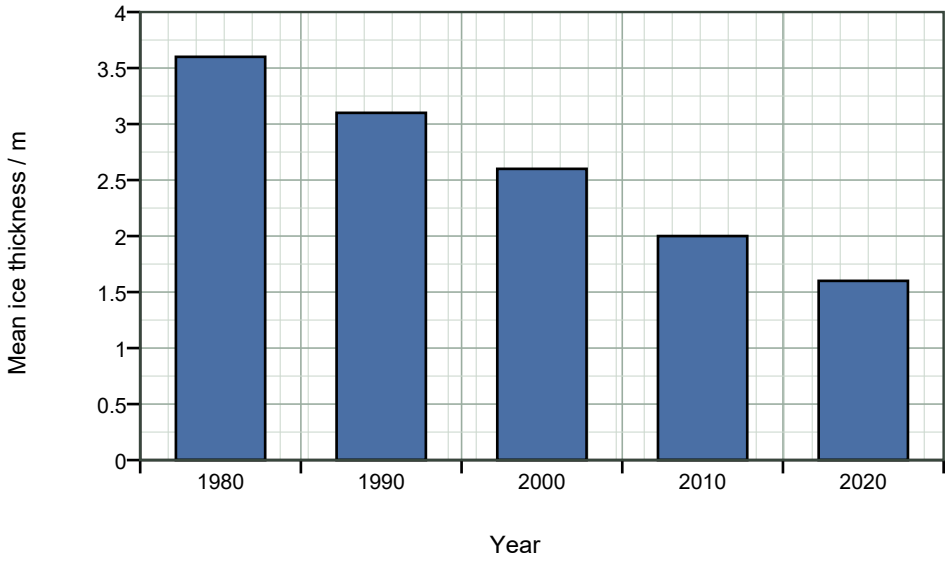
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0 2 Table 1 shows the area of Arctic sea ice in September (the annual minimum) in selected years. Figure 1 shows the mean thickness of Arctic sea ice in the same years.

Table 1

Year	1980	1990	2000	2010	2020
September sea ice area / million km^2	7.7	6.2	6.3	4.9	3.9

Figure 1



0 2 . 1 Calculate the percentage decrease in September sea ice area between 1980 and 2020.

[2 marks]

Answer ____ %

0 2 . 2 Explain why the loss of sea ice does **not** directly raise sea level, but the loss of land ice does.

[2 marks]

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0 2 . 3 Explain how the changes shown in Table 1 and Figure 1 could lead to further warming.

[3 marks]

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0 2 . 4 Explain **two** ways in which rising temperatures cause sea level to rise.

[2 marks]

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0 2 . 5 Explain why the break-up of an ice shelf can increase the rate at which land ice flows into the sea.

[1 mark]

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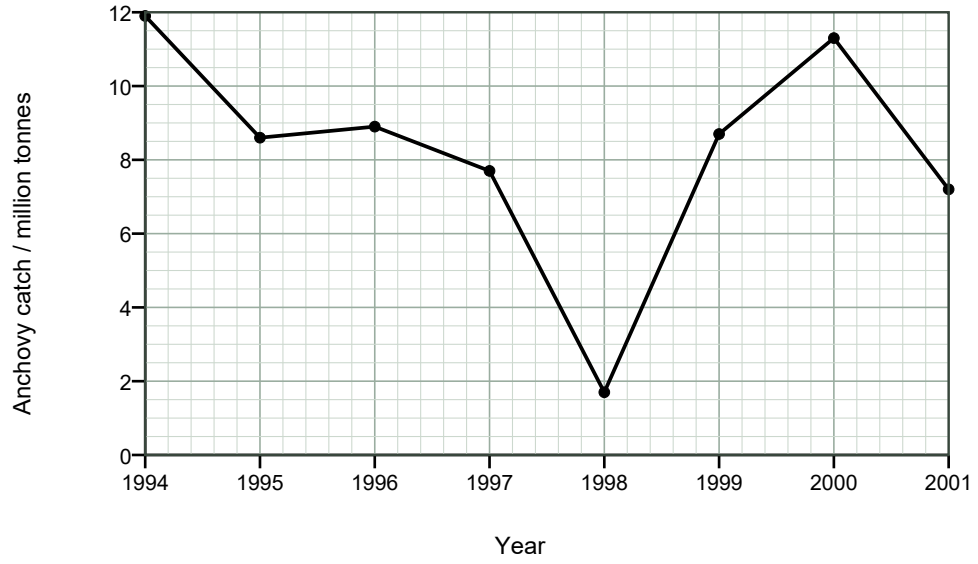
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0 3 Figure 1 shows the sea surface temperature anomaly in the eastern Pacific off Peru and the annual Peruvian anchovy catch for a series of years including a strong El Niño event in 1997-98.

Figure 1



0 3 . 1 Describe the normal (non-El Niño) pattern of winds and ocean currents in the tropical Pacific and explain why the Peruvian coast supports a rich fishery.

[4 marks]

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0 3 . 2 Explain how an El Niño event caused the collapse of the anchovy catch in 1998 shown in Figure 1.

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0 3 . 3 Describe **two** other effects of El Niño events.

[2 marks]

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0 3 . 4 Suggest why some scientists link global climate change to changes in the frequency of El Niño events, and why this link is uncertain.

[1 mark]

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0 4 . 1 Describe the two processes that drive the North Atlantic Conveyor.

[2 marks]

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0 4 . 2 Explain how global climate change could weaken the North Atlantic Conveyor and describe the possible consequence for the climate of the UK.

[4 marks]

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0 4 . 3 Explain why the effects of climate change on UK temperatures are difficult to predict.

[3 marks]

0 4 . 4 Describe how deep ocean currents are monitored.

[1 mark]

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0 5 . 1 Explain why it is difficult to be certain that a particular flood or heatwave is caused by anthropogenic climate change.

[3 marks]

0 5 . 2 Explain why historic temperature records collected in towns may overestimate global warming.

[2 marks]

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0 5 . 3 Describe how each of the following sources of proxy data is used to estimate past climates and give one limitation.

[5 marks]

Tree rings (dendrochronology)

Ice cores

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0 6 Table 1 compares two types of satellite used to monitor the Earth.

Table 1

Feature	Low Earth Orbit (polar) satellite	Geostationary satellite
Altitude	About 800 km	36 000 km above the equator
Orbit period	About 1.5 hours	24 hours (fixed position)
Coverage	Whole Earth surveyed in about 15 days	Same third of the Earth continuously
Detail	High resolution	Lower resolution

0 6 . 1 Use Table 1 to explain why LEO satellites are used to monitor changes in ice cover and vegetation, whereas geostationary satellites are used to monitor weather.

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0 6 . 2 Give **three** climate-related factors, other than ice cover, that are measured by satellite sensors.

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0 6 . 3 Explain how a computer climate model can be tested for reliability before its predictions are used.

[2 marks]

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0 6 . 4 Explain why representative data are essential for climate modelling.

[2 marks]

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0 7 . 1 Explain the difference between a negative and a positive feedback mechanism.

[2 marks]

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0 7 . 2 Describe **two** negative feedback mechanisms that may reduce global warming.

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0 7 . 3 Explain how melting permafrost and melting methane hydrate act as positive feedback mechanisms.

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0 7 . 4 Give **three** other positive feedback mechanisms.

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0 7 . 5 Explain what is meant by a climate **tipping point**.

[2 marks]

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0 8 . 1 Explain how climate change may affect the survival of species. Use named examples.

[6 marks]

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0 8 . 2 Explain why a species living in an interdependent community may be unable to move to a new area even if suitable conditions exist there.

[2 marks]

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0 8 . 3 Explain why polar regions warming faster than the equator affects the weather in the UK.

[2 marks]

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END OF QUESTIONS